Hints: Number theory – 4

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Every prime $p \in (m+1, 2m+1]$ divides $\binom{2m+1}{m} \leqslant 4^m$. So $\prod_{m+1<p\leqslant 2m+1} p \leqslant 4^m$. Induct: even n is not prime, and for odd $n = 2m+1$ use the bound up to $m+1$.

Hint 2. Bertrand: suppose there is no prime in $(n, 2n]$. By Kummer, $p^{v_p} \leqslant 2n$ for every p, so primes $p > \sqrt{2n}$ appear at most to the first power, and primes in $(2n/3, n]$ do not divide $\binom{2n}{n}$ at all.

Hint 3. Then $\frac{4^n}{2n+1} \leqslant \binom{2n}{n} \leqslant (2n)^{\sqrt{2n}} \cdot 4^{2n/3}$. This fails for n large (about $n \geqslant 500$). For smaller n, use the chain of primes $2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631$.

Problem 2

Hint 1. Kummer: a carry in base p at position j needs $p^{j+1} \leqslant$ the sum, so $v_p\binom{2n}{n} \leqslant \#\{j \geqslant 1 : p^j \leqslant 2n\} = v_p(\operatorname{lcm}(1, \ldots, 2n))$. Next, $(n+1)\binom{n}{k} = (k+1)\binom{n+1}{k+1}$. Let $p^e \leqslant n+1 < p^{e+1}$. Adding $k+1$ and $n-k$, no carry occurs below position $v_p(k+1)$ (the digits of $k+1$ are 0 there) and none out of position e or higher. So $v_p\big((k+1)\binom{n+1}{k+1}\big) \leqslant v_p(k+1) + (e - v_p(k+1)) = e$.

Hint 2. Lower bound: the middle coefficient is the largest of the $n+1$ numbers $\binom{n}{k}$, which sum to 2^n, so $\operatorname{lcm}(1, \ldots, n+1) \geqslant (n+1)\binom{n}{\lfloor n/2 \rfloor} \geqslant 2^n$. Third divisibility: by Legendre, $v_p\binom{2m}{m} = \sum_j (\lfloor 2m/p^j \rfloor - 2\lfloor m/p^j \rfloor)$, each term is $\geqslant 0$, and it equals 1 when $m < p^j \leqslant 2m$. Hence $v_p\binom{2m}{m} \geqslant v_p \operatorname{lcm}(1..2m) - v_p \operatorname{lcm}(1..m)$. In the same way $\operatorname{lcm}(1, \ldots, 2m+1) \mid \operatorname{lcm}(1, \ldots, m+1)\binom{2m+1}{m}$.

Hint 3. Upper bound: prove $L(N) = \operatorname{lcm}(1, \ldots, N) \leqslant 4^{N-1}$ by strong induction ($L(1) = 1$, $L(2) = 2$): $L(2m) \leqslant L(m)\binom{2m}{m} \leqslant 4^{m-1}4^m$ and $L(2m+1) \leqslant L(m+1)\binom{2m+1}{m} \leqslant 4^m \cdot 4^m$, using $\binom{2m+1}{m} \leqslant \frac{1}{2}2^{2m+1}$. Primes: $L(2n) \leqslant (2n)^{\pi(2n)}$ (each $p^{v_p} \leqslant 2n$) and $L(2n) \geqslant \binom{2n}{n} \geqslant \frac{4^n}{2n+1}$, so $\pi(2n) \geqslant \frac{n \log 4 - \log(2n+1)}{\log 2n}$. (The sharper $\log L(n) \sim n$ is equivalent to the prime number theorem.)

Problem 3

Hint 1. $v_2\binom{2n}{n} = s_2(n)$ (Kummer: doubling carries once per digit 1). So $v_2 = 1$ exactly when n is a power of 2.

Hint 2. Since $s_2(n) \geqslant 1$, squarefree forces $v_2 = 1$, hence $n = 2^m$. For $n \leqslant 100$ only $1, 2, 4, 8, 16, 32, 64$ remain; $n = 1, 2, 4$ give $2, 6, 70$, which are squarefree.

Hint 3. Look at other primes with Kummer. In base 3: $8 = 22_3$, $16 = 121_3$, $32 = 1012_3$, and doubling each of them carries twice, so $9 \mid \binom{2n}{n}$. For $64 = 2101_3$ there is only one carry, but $64 = 224_5$ and doubling carries three times, so $125 \mid \binom{128}{64}$. Answer: $n \in \{1, 2, 4\}$. (These are in fact the only n at all, but that is a deep theorem of Granville and Ramaré.)

Problem 4

Hint 1. Work modulo powers of p with fractions whose denominators are prime to p. Pair k with $p-k$: $\frac{1}{k} + \frac{1}{p-k} = \frac{p}{k(p-k)}$, so $H_{p-1} = \frac{p}{2}\sum_{k=1}^{p-1} \frac{1}{k(p-k)}$, and $\frac{1}{k(p-k)} \equiv -\frac{1}{k^2} \pmod p$.

Hint 2. $k \mapsto k^{-1}$ permutes the non-zero residues, so $\sum_{k=1}^{p-1} k^{-2} \equiv \sum k^2 = \frac{(p-1)p(2p-1)}{6} \equiv 0 \pmod p$ for $p \geqslant 5$. This gives $v_p\big(\sum k^{-2}\big) \geqslant 1$, and then $v_p(H_{p-1}) \geqslant 1 + 1 = 2$.

Hint 3. For $p = 3$: $1 + \frac{1}{2} = \frac{3}{2}$ has $v_3 = 1$, and $1 + \frac{1}{4} = \frac{5}{4}$ has $v_3 = 0$ (the step $\sum k^2 \equiv 0$ needs $p \geqslant 5$). Remark: the bounds are usually sharp, with equality failing only for the Wolstenholme primes 16843, 2124679, ...; so there is no closed formula for these valuations.

Problem 5

Hint 1. Odd n: take $k = n^m$. Then $a_k = n^{n^m - m}$ is odd.

Hint 2. Even n: find k with $k \mid n^k - r$ for some small r with n^k/k just above $(n^k - r)/k$. Try $k = p \cdot 2^m$, where p is an odd prime factor of $n^{2^m} - 2^m$ (it exists, since $n^{2^m} - 2^m = 2^m(\text{odd} > 1)$).

Hint 3. By Fermat, $n^k = (n^{2^m})^p \equiv (2^m)^p \equiv 2^m \pmod{p}$. Since $0 < 2^m < k$, we get $a_k = \frac{n^k - 2^m}{p\,2^m}$, which is odd because $n^k/2^m$ is even.

Problem 6

Hint 1. Check that $a = 1, 3, 5$ work ($n = 1, 1, 4$). For even $a = 2^\alpha d$, pick $i < a$ with $n + i = 2^{\alpha-1} e$, e odd. Then $t(n + i + a) = 2d + e \equiv e + 2 \pmod{4}$.

Hint 2. Odd $a > 8$: some $n + i = 2d$ (d odd) with $i \leqslant a - 5$. Then $t(n + i)$ and $t(n + i + 4)$ differ by 2 mod 4, while $n + a + i$ and $n + a + i + 4$ are odd and congruent mod 4. So both differences cannot vanish mod 4.

Hint 3. $a = 7$: some $n + i \in \{8k + 3, 8k + 6\}$, and both cases fail directly. Answer: $a \in \{1, 3, 5\}$.

Problem 7

Hint 1. If $n = \prod p_i^{e_i}$ has 2^k divisors, then each $e_i + 1$ is a power of 2. So n is a product of distinct numbers of the form p^{2^j} ("Fermi–Dirac primes"), exactly k of them.

Hint 2. Minimality: $f(2^k)$ is the product of the k smallest elements of $\{p^{2^j}\}$. (If a smaller unused element exists, swap it for a larger one; check that the result is still of the right shape.)

Hint 3. Hence $f(2^{k+1}) = f(2^k) \cdot (\text{the next element})$.

Problem 8

Hint 1. Bertrand: there is a prime p with $n < p \leqslant 2n$ (in fact $p < 2n$ for $n > 1$).

Hint 2. By Kummer or directly, $v_p\binom{2n}{n} = 1$ (p divides $(2n)!$ once and $n!$ not at all). So $\binom{2n}{n}$ is not a perfect power.

Problem 9

Hint 1. For a fixed prime p, let $p^e \leqslant n + 1 < p^{e+1}$. Show that $\max_k v_p\binom{n}{k} = e - v_p(n + 1)$. By Kummer it is the maximal number of carries in $k + (n - k)$.

Hint 2. Let $t = v_p(n+1)$, the number of trailing digits $p-1$ of n. Show that no carry can occur in these t lowest positions (induct from position 0). Also no carry occurs at position e or above. So there are at most $e - t$ carries. Take $k = p^e - 1 \leqslant n$: then $n - k = n + 1 - p^e$, and the addition carries at every position $t, \ldots, e - 1$. Comparing with $v_p(\operatorname{lcm}(1, \ldots, n + 1)/(n + 1)) = e - t$ proves the identity.

Hint 3. For the sum, use $\frac{1}{\binom{n}{k}} = (n + 1) \int_0^1 t^k (1 - t)^{n-k}\, dt$. Summing the geometric series under the integral gives $\sum_k \frac{(-1)^k}{\binom{n}{k}} = \frac{(n+1)(1+(-1)^n)}{n+2}$.

Problem 10

Hint 1. Legendre: $v_p\left(\frac{n!}{k_1!\cdots k_m!}\right) = \frac{\sum s_p(k_i) - s_p(n)}{p-1}$.

Hint 2. Adding $k_1, \ldots, k_m$ column by column, each carry of size c (an amount c carried to the next position) lowers the total digit sum by $c(p-1)$. So the valuation is the total amount carried.

Hint 3. The multinomial coefficient is coprime to p exactly when, in every base-p position, the digits of the k_i sum to the digit of n (no carries).